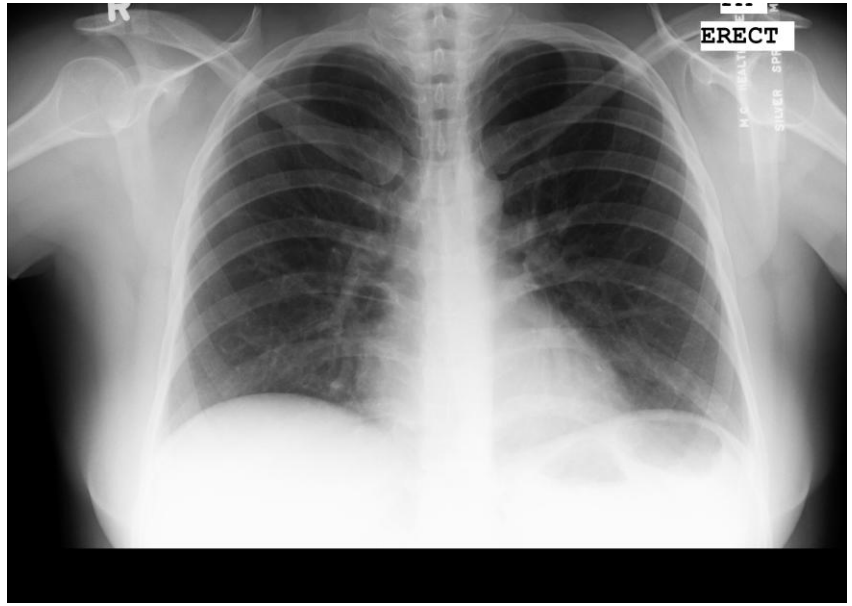
BASE MODEL

CXR benchmark reproduction

Source-only UNet before test-time adaptation

SZ-CXR → Montgomery

No source augmentation | Seed 42 | 138 target cases



Target Dice

0.9396

Base checkpoint before TTA

Paper target specification

The base model must match the shared UNet setup before comparing adaptation methods.

Component	Paper specification	Formal local run
Task	SZ-CXR lung segmentation → Montgomery	SZ-CXR lung segmentation → Montgomery
Input	256 x 256, grayscale	256 x 256, per-image z-score
Output	2 channels, binary foreground	2-channel UNet head
Encoder	5 blocks, base width C0=32	[32, 64, 128, 256, 512]
Split	Source 8:2, full target test	453 train / 113 validation / 138 target
Data	662 source images	566 exact image-mask pairs

Completeness result

No audited public candidate supplied masks for all 662 CHNCXR IDs.

Two relevant Shenzhen mask archives both contain 566 exact pairs.

Base-model training flow

The source-only checkpoint uses no source augmentation.

1. **Load paired source**
   566 exact image-mask pairs
2. **Split source**
   453 train / 113 validation
3. **Normalize**
   Resize 256 x 256; per-image **z-score**
4. **Optimize UNet**
   Adam, learning rate 0.001, batch 2, no augmentation
5. **Select checkpoint**
   Best validation Dice: 0.9562 at epoch 14, Early stopping patience：5
6. **Evaluate target**
   138 Montgomery images, no TTA update

Checkpoint contract

Input: image only

Target masks are excluded from training, validation and prediction.

The same checkpoint is copied for later TENT, TestFit and GraTa runs.

Source-only means no parameter update at target time.

Base-model result

The checkpoint transfers to Montgomery before any test-time update.



138 target cases

Dice

0.9396

HD95

2.64 px

JI 0.8903 | Recall 0.8971 | Precision 0.9920

Paper baseline is shown for context only. The source-mask coverage and exact training recipe differ.

Reproducibility record

Field	Recorded value
Checkpoint	source_checkpoint.pth
SHA-256	d08d23748329fdfaef5842c6257ee418b55d2ec8a083790c9c8b2eff3d67cfb2
Best source validation	Dice 0.9562 at epoch 14; 19 epochs executed
Target evaluation	138 predictions and 138 metric rows
Runtime	Windows 11 | PyTorch 2.11.0+cu128 | NVIDIA RTX 5060 Laptop GPU
Metric policy	Vendor-compatible all-foreground HD95; 256 x 256 pixel units; population SD

Re-run

```
.\.venv\Scripts\python.exe -m experiments.feature_alignment_2d_pilot.run_base_model
```

Base model conclusion

What the base run establishes

- The formal base uses no source augmentation.
- The source checkpoint reaches 0.9562 validation Dice.
- It reaches 0.9396 Dice on all 138 Montgomery cases before TTA.
- Target masks stay outside training and inference.

What remains unresolved

- The public masks cover 566 of 662 source images.
- The paper's complete processed source archive is not public in the audited candidates.
- The local UNet training recipe is evidence-bounded, not a line-by-line official source-training reproduction.
- Multi-seed stability remains to be evaluated for this no-augmentation base.

Base model first. TTA comparisons are meaningful only after this checkpoint is fixed.

Group Member	Contribution	Work Duty